

Basic Information

Experiment No.: 06

Date: 24-09-2026

Title: Java Iterator Interface using ArrayList

Objective / Aim

To implement the Iterator interface to traverse elements of an "ArrayList" and remove a specific element safely during iteration.

System Requirements

Hardware

- Computer/Laptop
- Minimum 4 GB RAM

Software

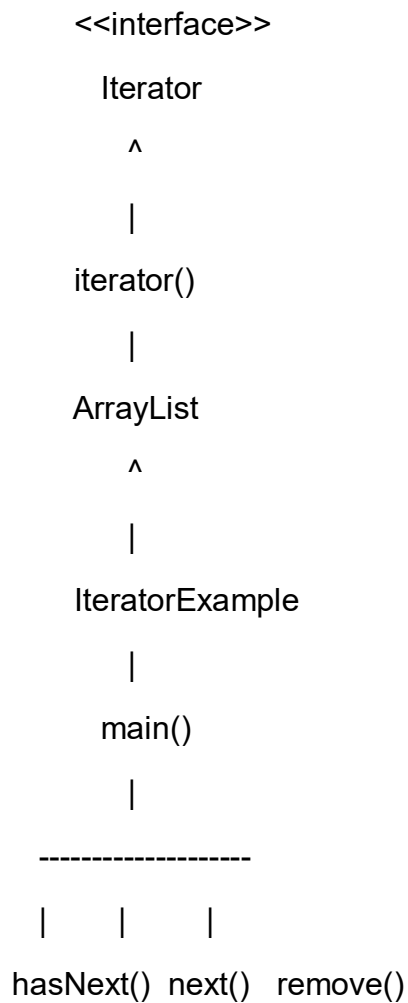
- Java JDK
 - Any Java IDE or Command Prompt
 - Operating System: Windows/Linux/macOS
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Algorithm / Logic Description

1. Create an "ArrayList" and add elements.
2. Create an "Iterator" using "list.iterator()".
3. Check whether the next element exists using "hasNext()".
4. Read the element using "next()".
5. Display the element.

6. Check whether the element is ""Banana"".
 7. Remove ""Banana"" using "it.remove()".
 8. Display the list after removal.
 9. End the program.
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UML Class Diagram



Source Code

```
package collectionframework;
```

```
import java.util.Iterator;
import java.util.List;
import java.util.ArrayList;

public class IteratorExample {
    public static void main(String[] args) {

        List<String> list = new ArrayList<>();

        list.add("Apple");
        list.add("Banana");
        list.add("Mango");

        Iterator<String> it = list.iterator();

        while (it.hasNext()) {

            String item = it.next();

            System.out.println(item);

            // Remove Banana
            if (item.equals("Banana")) {
                it.remove();
            }
        }

        System.out.println("After removal: " + list);
    }
}
```

```
}  
}
```

Compilation & Execution Commands

Save the file as:

IteratorExample.java

Since the program contains:

package collectionframework;

the file should be inside the "collectionframework" folder.

Compile:

```
javac IteratorExample.java
```

Run:

```
java collectionframework.IteratorExample
```

Output Screen

Apple

Banana

Mango

After removal: [Apple, Mango]

Result

The Iterator interface was successfully used to traverse the "ArrayList" and safely remove the ""Banana"" element during iteration.